

### c) Inter-process communication

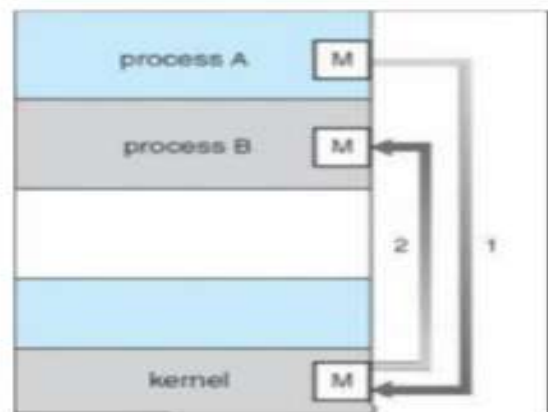
When processes share data among themselves, data can be some kind of information, the communication which is established is known as Inter-process communication.

Two co-operative processes communicate with each other known as Inter-process communication.

There are two models or types of Inter-process communication.

1. Shared memory.
2. Message Passing.

#### 2. Message Passing



Processes exchange messages to communicate.

Useful when processes are on different computers in a network.

Processes do not share memory; they use the kernel to send and receive messages.

There must be a communication link between the processes, with one link for each pair of communicating processes.

## Shared Memory

In this process create a specific memory region to communicate.

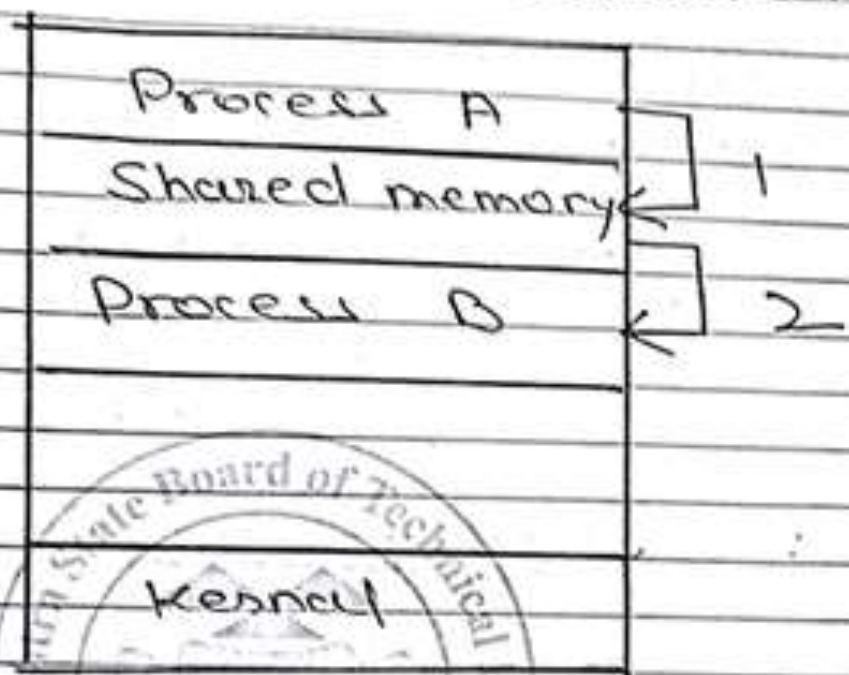


Fig: Shared Memory

In this mechanism, both the co-operative process share some region which is termed as "shared memory" where they can write and read data.

In this region, information and data is ~~stop~~ shared but kernel is ~~an~~ unaware about it.

Process exchange information among themselves without use of O.S (kernel) is Shared Memory

### Advantages

- ① Simple to Implement
- ② Effective in networks.

### DisAdvantages

- ① Can't share data or info. when processes resides in different systems.
- ② Faster Once but as compared to message passing security issue.

Message passing advantages disadvantages

### Advantage

- most Secured Mechanism
- Slower - But Effective.

### DisAdvantages

- slower in speed.
- there is intermediate message passing.

Homework : compare message passing vs shared memory

**Explain working of CPU switch from process to process with neat labelled diagram.**

4M

A CPU switch from process to process is referred as **context switch**.

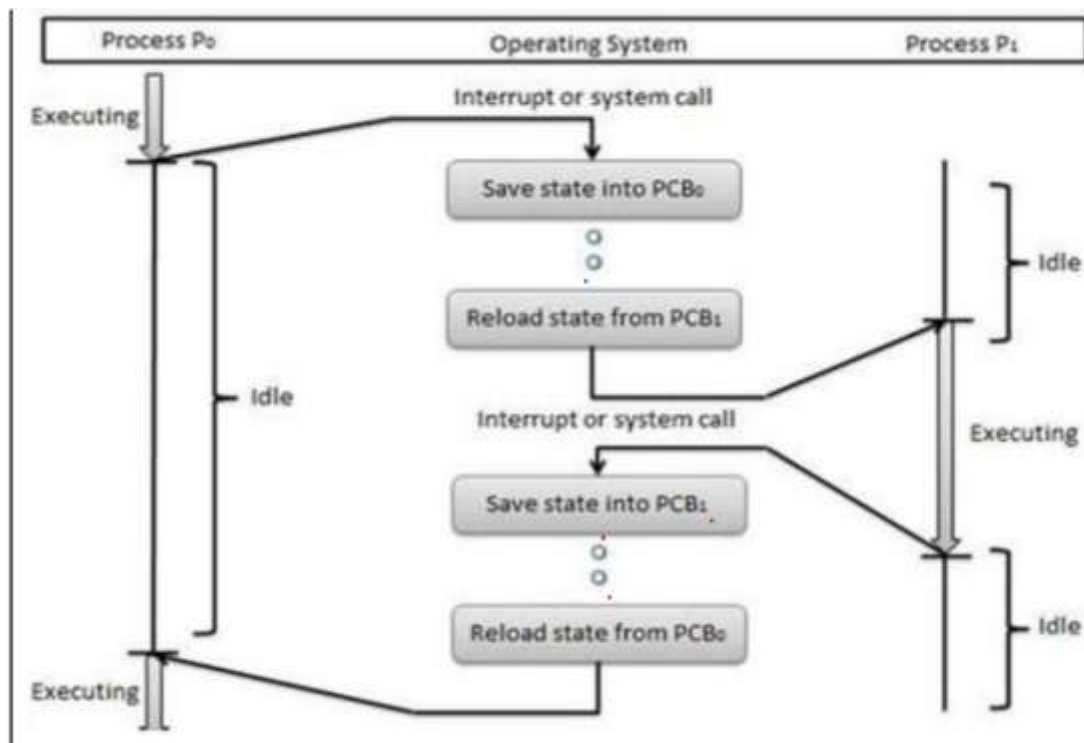
Explanation  
n  
2M

A context switch is a mechanism that store and restore the state or context of a CPU in Process Control block so that a process execution can be resumed from the same point at a later time. When the scheduler switches the CPU from one process to another process, the context switch saves the contents of all process registers for the process being removed from the CPU, in its process control block.

Diagram  
2M

Context switch includes two operations such as state save and state restore. State save operation stores the current information of running process into its PCB. State restore operation restores the information of process to be executed from its PCB. Switching the CPU from one process to another process requires performing state save operation for the currently executing process (blocked) and a state restore operation for the process ready for execution. This task is known as context switch.

Relevant  
Explanation  
n shall be  
considered



### 1. top

- Shows **real-time list** of running processes and their resource usage (CPU, memory).
- Example:

```
bash
```

[Copy code](#)

```
top
```

```
top
```

Output (sample):

```
sql
```

[Copy code](#)

```
top - 12:03:11 up 1 day, 3:11, 2 users, load average: 0.01, 0.05, 0.08
PID   USER     PR  NI  VIRT  RES  SHR S %CPU %MEM    TIME+  COMMAND
1011  root      20   0 392m   8m   4m S   0.0   0.2   0:01  bash
...
```

### 3. kill

- **Stops** a process by ID (PID).
- Example:

```
bash
```

[Copy code](#)

```
kill 1234      # Kill process with PID 1234
kill -9 1234   # Force kill
```

iv) kill

It stops the execution of the running process.

It state that given is running of process it should running that process and exit/terminate and allowed for garbage collection of memory.

6 Syntax: \$ kill pid

Example: \$ kill 12345

Kills the process 12345.

There are also options available

#### 4. wait

- Makes the shell wait until a specific process finishes.
- Example:

```
bash
```

[Copy code](#)

```
wait 1234 # Wait for process 1234 to finish
```

ii) wait

→ If the process is in running state it takes that process into waiting state by specifying its PID (Process Identification Number).

Syntax : \$wait PID

Example : \$wait 12345

#### 5. `sleep`

- Pauses execution for given time (seconds).
- Example:

```
bash
```

```
sleep 5    # Wait for 5 seconds
```

### iii) sleep

It stops the terminal for specified time given by the user.

Syntax: `$sleep time [unit]`.

time should be specified in seconds.

### Example

`$sleep 15`

It will stop terminal for 15 seconds.

#### 6. exit Ask ChatGPT

- Ends the current shell or script.
- Example:

```
bash
```

[Copy code](#)

```
exit
```

#### 7. nice

- Runs a process with a priority level (lower priority = higher "nice" value).
- Example:

```
bash
```

[Copy code](#)

```
nice -n 10 myprogram # Run myprogram with nice value 10
```

a)

i) ps

⇒ It list the process characteristics.

Syntax : \$ps - - -

| <u>Options</u> | <u>Syntax</u>                                       | <u>Description</u>                                      |
|----------------|-----------------------------------------------------|---------------------------------------------------------|
| -u             | \$ps -u <u>ABC</u><br>or<br>\$ps -u <u>username</u> | It list the processes <u>related</u> to specified user. |
| -a             | \$ps -a                                             | It list the processes of all users.                     |
| -e             | \$ps -e                                             | list all the process with cmd user.                     |
| -l             | \$ps -l                                             | list all commands with PPID, C etc.                     |

Example : \$ps

| PID   | Time     | TTYL  | cmd  |
|-------|----------|-------|------|
| 12345 | 00:00:00 | pts/0 | bash |
| 12512 | 00:00:00 | pts/0 | bash |

P.T.O

|            |                                                                                                                                                                                                                                            |                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>a</b>   | <b>Write Unix command for following:</b><br>i) create a folder OSY    ii) create a file FIRST in OSY folder<br>iii) List/display all files and directories.<br>iv) Write command to clear the screen                                       | <b>4M</b>                      |
| <b>Ans</b> | i) create a folder OSY:<br>\$mkdir OSY<br><br>ii) create a file FIRST in OSY folder:<br>\$cd OSY<br>\$cat>FIRST    or    \$ touch FIRST<br><br>iii) List/display all files and directories:<br>\$ls<br><br>iv) to clear screen:<br>\$clear | <b>Each correct command-1M</b> |

|             |                                                                                                                                                                                                                                                                                                           |                                   |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <b>b)</b>   | <b>Writer the outputs of following commands</b><br>(i) Wait 2385018<br>(ii) Sleep 09<br>(iii) PS -u Asha                                                                                                                                                                                                  | <b>6M</b>                         |
| <b>Ans.</b> | i) Wait command waits until the termination of specified process ID 2385018<br>ii) Sleep command is used to delay for 9 seconds during the execution of a process i.e. it will pause the terminal for 9 seconds.<br>iii) ps command with -u is used to display data/processes for the specific user Asha. | <b>2M for each correct output</b> |

f) Give commands to perform following tasks:

- i) To add delay in script
- ii) To terminate a process

### Unix Command Questions

- Create a folder named DATA and inside it create another folder BACKUP.
- Copy a file report.txt into the BACKUP folder.
- Move the file report.txt from BACKUP back to the DATA folder.
- Delete the file report.txt from DATA folder.

5. Display the current working directory.
6. Display the date and time.
7. Show who is currently logged in.
8. Display the first 10 lines of the file sample.txt.

9. Rename a file oldfile.txt to newfile.txt.
10. Count the number of words in a file essay.txt.
11. Count the number of lines in a file essay.txt.
12. Display contents of a file notes.txt page by page.

13. Find all files in the current directory.
14. Search the word Linux in file book.txt.
15. Sort the contents of a file marks.txt.
16. Display only unique lines from file data.txt.

17. Give command to change permissions of file script.sh to make it executable.
18. Show hidden files in the current directory.
19. Kill a process with process ID 1234.
20. Log out from the terminal.